

Synthesis of Multisubstituted 1,4-Dihydrobenzoxazin-2-ones through a One-Pot Nucleophilic N-Alkylation/C-Alkylation of Cyclic α -Imino Esters

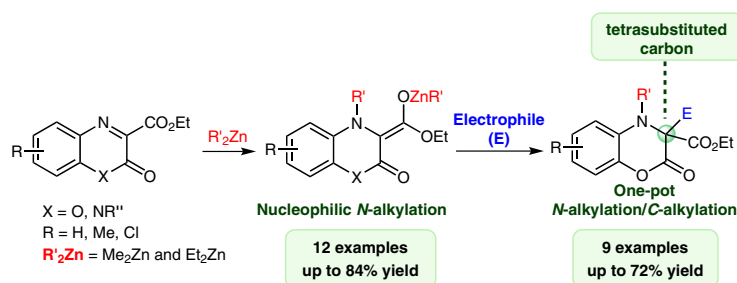
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Abstract A nucleophilic N-alkylation of 2-oxobenzoxazine-2-carboxylates with organozinc reagents with good selectivities and in moderate to good yields is described. Moreover, the synthesis of multisubstituted 1,4-dihydrobenzoxazine-2-ones bearing a tetrasubstituted carbon atom via a one-pot N-alkylation/C-alkylation reactions is reported.

Key words dihydrobenzoxazinone, umpolung, zinc, tetrasubstituted carbon, one-pot reaction, cyclic α -imino esters

1,4-Benzoxazine derivatives are an important class of heterocyclic compounds present in numerous natural products, which have played crucial roles in the area of medicinal chemistry over the past decades because of their widespread range of biological and pharmaceutical activities.¹ Examples of dihydrobenzoxazine derivatives with biological activities are the compound **A** with antibacterial properties² and the compound **B** that shows hypolipidemic, antiobesity, and hypocholesterolemic activities.³ A related scaffold is 1,4-quinoxaline, which also exhibits important biological activities.⁴ Thus, compound **C** shows antiviral activity⁵ and compound **D** presents antitumor activity⁶ (Fig-

ure 1). Consequently, the synthesis of new 1,4-benzoxazine (and also 1,4-quinoxaline) derivatives has been and continues to be attracting the attention of synthetic organic chemists and medicinal chemists.

Traditional methods for building the framework of 1,4-benzoxazines and 1,4-quinoxalines use 2-nitrophenols, 2-aminophenols, or benzene-1,2-diamines as starting materials,^{1b,7} although recently several transition-metal-catalyzed reactions have been described.⁸ Alternatively, the modification of substituents on the 1,4-benzoxazines have been carried out, particularly by the addition of nucleophilic reagents to cyclic imines⁹ (reaction 1, Scheme 1). We envisioned that the installation of one additional electron-withdrawing group at C-3 position as in **2** (Scheme 1) will lead to a highly polarized C=N bond,^{10,11} which could be considered as an efficient way to induce umpolung of the imine functionality. However, considering the system as a whole, it would be an α,β -unsaturated ester that would undergo conjugate addition at the nitrogen atom of the imine functionality. In any case, it would be a nucleophilic alkylation at the nitrogen atom of the imine group. Moreover, the intermediate metal enolate obtained in this nucleophilic N-alkylation reaction could experience a subsequent C–C bond formation by reaction with an alkylating reagent. In

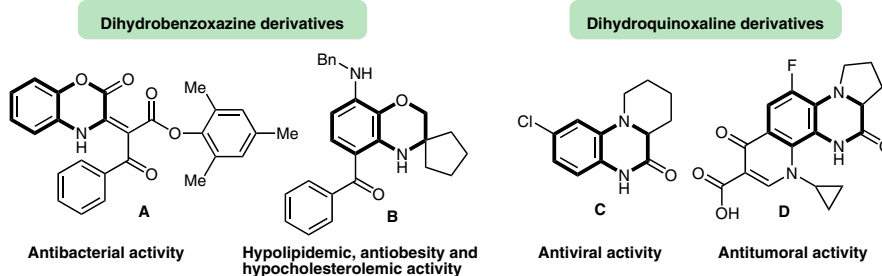
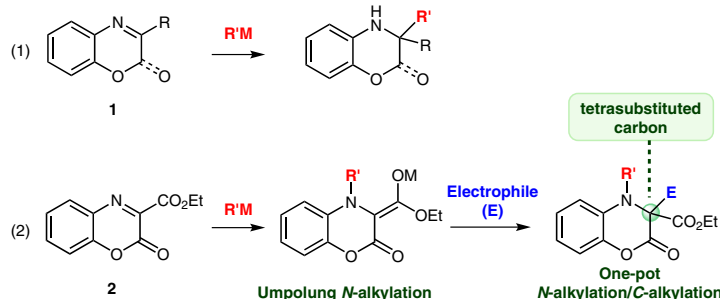


Figure 1 Biologically active dihydrobenzoxazine and dihydroquinoxaline derivatives



Scheme 1 Organometallic additions to benzoxazine derivatives

the literature,^{12–15} several successful examples for the regioselective N-alkylation of acyclic α -imino esters have been described by the groups of Shimizu,^{13a–d,14a–e} Kozłowski,^{13e–g} and others.¹⁵ However, the N-alkylation of the cyclic α -imino esters such as **2** has been scarcely studied, despite its synthetic potential.^{16,17} Herein, we present our efforts toward the nucleophilic N-alkylation of 2-oxobenzoxazine-2-carboxylates with organozinc reagents. Furthermore, we report the synthesis of multisubstituted 1,4-dihydrobenzoxazin-2-ones bearing a quaternary carbon atom by a one-pot N-alkylation followed by a C-alkylation of the enolate produced with alkyl halides.

Our studies began with the optimization of the reaction conditions for the N-alkylation step, using ethyl 2-oxo-2H-benzo[*b*][1,4]oxazine-3-carboxylate (**2a**) as a model substrate and Me_2Zn (**3a**) as the organozinc reagent. As shown in Table 1, the optimization process was started with the screening of various solvents (CH_2Cl_2 , toluene, and THF, Table 1, entries 1–3) at room temperature, obtaining the best yield of the N-alkylation product **4aa** when CH_2Cl_2 was used. Next, different temperatures for the Me_2Zn addition reaction were evaluated. The best yield of product **4aa** (66%) was obtained when Me_2Zn was added at 0 °C and then the reaction mixture was allowed to reach room temperature (entry 4).

Under the best reaction conditions (Table 1, entry 4), the scope of the reaction was evaluated using differently substituted ethyl 2-oxobenzoxazine-3-carboxylates **2** (Scheme 2). Electron-donating groups at the 6 or 7 position of the arene ring of the benzoxazine were well tolerated and the corresponding N-alkylated products **4ba** and **4da** were obtained in good yields. However, when a chlorine atom was installed at position 6 or 7, the corresponding product was obtained in lower yield. When the reaction was quenched with MeOH, a transesterification reaction was observed and the corresponding methyl 4-methyl-2-oxo-3,4-dihydro-2H-benzo[*b*][1,4]oxazine-3-carboxylate (**4aa'**) was obtained in 56% yield. After the study of the N-alkylation with Me_2Zn (**3a**), the reaction was tested with Et_2Zn (**3b**). The corresponding N-alkylation products **4ab**, **4cb**, and **4db** were obtained in slightly better yields than when Me_2Zn was used as organometallic reagent. Very re-

cently, Shimizu has developed the N-alkylation of quinoxaline derivatives with Grignard reagents.¹⁶ Therefore, we also decided to examine the addition of dialkylzinc reagents to ethyl 3-oxo-3,4-dihydroquinoxaline-2-carboxylate derivatives **6** bearing different N-protecting groups such as methyl, benzyl, or $\text{CH}_2\text{CO}_2\text{Me}$. When the reaction was performed with Et_2Zn ,¹⁸ the desired N-alkylated products **7ab**, **7bb**, and **7cb** were obtained in good yields (71–84%).

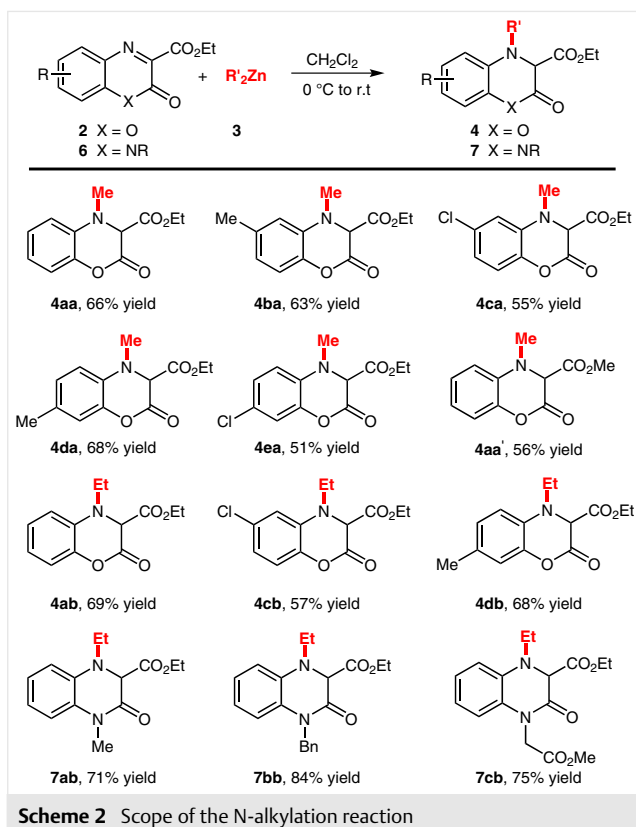
To increase the synthetic applicability of the N-alkylation of benzoxazine and quinoxaline derivatives, the alkylation of the enolates formed after the initial addition of the organozinc reagent was attempted. In this manner, a tandem N-alkylation/C-alkylation reaction would be possible for the one-pot three-component coupling reaction providing dihydrobenzoxazine derivatives with a tetrasubstituted carbon atom. For this purpose, after the addition of the organozinc reagent, benzyl bromide or allyl bromide (5 equiv) were added to quench the reaction mixture. The corresponding enolate generated after the addition of Et_2Zn was found to be more reactive than the one formed from Me_2Zn . The corresponding Et_2Zn enolate was trapped with benzyl

Table 1 Optimization of Reaction Conditions for the N-Alkylation Step^a

Entry	Me_2Zn (3a) (equiv)	Solvent	Temp (°C)	Yield (%) ^b of	
				4aa	5aa
1	2	CH_2Cl_2	r.t.	60	5
2	2	toluene	r.t.	45	9
3	2	THF	r.t.	36	–
4	2	CH_2Cl_2	0 to r.t.	66	10
5	2	CH_2Cl_2	–78 to r.t.	54	7

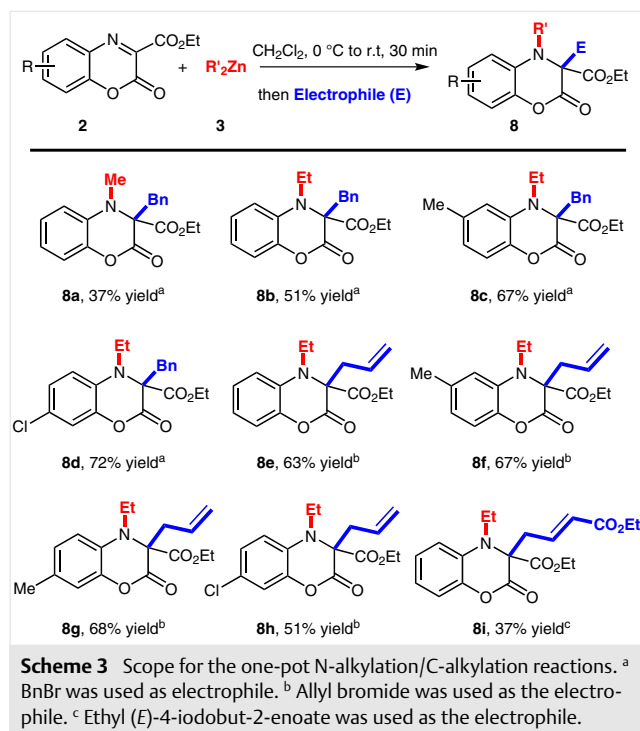
^a Reaction conditions: **2a** (0.15 mmol), **3a** (2 M in toluene, 0.30 mmol).

^b Yield after column chromatography.



bromide, affording the corresponding product **8b** in 51% yield. On the other hand, product **8a**, obtained from Me_2Zn enolate, was obtained in lower yield (37%). The corresponding N,C-dialkylated products **8c** and **8d**, with different substitution pattern in the aromatic ring of the benzoxazine scaffold were obtained in better yields (67% and 72%, respectively). Furthermore, allyl bromide was tested as electrophile for trapping the enolate. The corresponding allyl substituted products **8e–h** were obtained in good yields (Scheme 3). Our efforts for a tandem N-alkylation/C-alkylation reaction with quinoxaline derivatives were unsuccessful, due to the lack of reactivity of the corresponding zinc-quinoxaline enolate. Finally, when ethyl (*E*)-4-iodobut-2-enoate was tested as electrophile, poor conversion to compound **8i** was achieved. In order to increase the conversion, different additives were tried, but all the attempts failed. Therefore, the α,β -unsaturated ester **8i** with a benzoxazine skeleton, was synthesized from the corresponding lithium enolate prepared from compound **4ab** and LDA in THF, and trapped with ethyl (*E*)-4-iodobut-2-enoate.

In summary, we have developed a one-pot synthesis of multisubstituted dihydrobenzoxazin-2-ones using a tandem N-alkylation/C-alkylation process. The reaction proceeds through a nucleophilic addition of organozinc reagents to the nitrogen of the benzoxazine derivative. The resulting zinc enolate can be trapped with electrophiles to



form dihydrobenzoxazin-2-ones with a tetrasubstituted carbon atom in moderate to good yields. This methodology provides a straightforward procedure for the synthesis of a variety of pharmacologically and synthetically useful nitrogen heterocycles.

Reactions were carried out under N_2 in test tubes or round-bottomed flasks oven-dried overnight at 120 °C. CH_2Cl_2 and toluene were distilled from CaH_2 . THF was distilled from sodium benzophenone ketyl. Reactions were monitored by TLC analysis using Merck silica gel 60 F-254 thin layer plates. Flash column chromatography was performed on Merck silica gel 60, 0.040–0.063 mm. Melting points were determined in capillary tubes. NMR spectra were run at 300 MHz for ^1H and at 75 MHz for ^{13}C NMR using residual nondeuterated solvent as internal standard (CHCl_3 : 7.26 and 77.0 ppm). Chemical shifts are given in ppm. The carbon type was determined by DEPT experiments. High-resolution mass spectra (ESI) were recorded on a AB SCIEX Triple TOFTM spectrometer equipped with an electrospray source with a capillary voltage of 4.5 kV (ESI). Ethyl 2-oxo-2H-benzo[b][1,4]oxazine-3-carboxylates **2** and ethyl 3-oxo-3,4-dihydroquinoxaline-2-carboxylates **6** were prepared from the corresponding aminophenols or the corresponding benzene-1,2-diamines as described in the literature.^{9i,19}

Nucleophilic N-Alkylation of Benzoxazinones; General Procedure

To a solution of benzoxazinone **2** (0.15 mmol) in CH_2Cl_2 (2 mL) at 0 °C, was added dropwise a 2 M Me_2Zn solution in toluene/1 M Et_2Zn solution in hexanes **3** (0.3 mmol) under N_2 atmosphere. After the addition, the reaction mixture was placed at r.t. and stirred till completion (TLC, $\pm$ 30 min). The reaction was quenched with an aq solution of Rochelle salt (5 mL), extracted with CH_2Cl_2 (3 $\times$ 15 mL), the combined

organic layers were dried (anhyd MgSO₄), and concentrated under reduced pressure. The resulting product was purified by flash column chromatography on silica gel (hexane/EtOAc) to afford the desired product **4**.

Ethyl 4-Methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4aa)

Yield: 23.4 mg (66%); brown oil.

IR (neat): 1769, 1683, 1502, 1282, 1200, 1013, 745 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 7.18–7.00 (m, 2 H), 6.93–6.78 (m, 2 H), 4.65 (s, 1 H), 4.13 (q, *J* = 7.1 Hz, 2 H), 3.05 (s, 3 H), 1.16 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 165.43 (C), 160.55 (C), 141.16 (C), 133.45 (C), 125.52 (CH), 120.16 (CH), 116.48 (CH), 112.94 (CH), 64.94 (CH), 62.36 (CH₂), 36.43 (CH₃), 13.9 (CH₃).

HRMS (ESI⁺): *m/z* = 236.0918 [M + H]⁺, C₁₂H₁₄NO₄ requires 236.0923.

Ethyl 4,6-Dimethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4ba)

Yield: 23.6 mg (63%); green oil.

IR (neat): 1767, 1682, 1508, 1208, 1016, 750 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 6.92 (d, *J* = 8.1 Hz, 1 H), 6.71–6.58 (m, 2 H), 4.62 (s, 1 H), 4.13 (q, *J* = 7.1 Hz, 2 H), 3.03 (s, 3 H), 2.33 (s, 3 H), 1.17 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 165.55 (C), 160.63 (C), 139.15 (C), 135.29 (C), 133.09 (C), 120.58 (CH), 116.11 (CH), 113.52 (CH), 64.95 (CH), 62.30 (CH₂), 36.38 (CH₃), 21.29 (CH₃), 13.96 (CH₃).

HRMS (ESI⁺): *m/z* = 250.1076 [M + H]⁺, C₁₃H₁₆NO₄ requires 250.1079.

Ethyl 6-Chloro-4-methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4ca)

Yield: 22.3 mg (55%); orange oil.

IR (neat): 1777, 1728, 1500, 1202, 1036, 837, 806 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 6.96 (d, *J* = 8.5 Hz, 1 H), 6.84 (dd, *J* = 8.4, 2.3 Hz, 1 H), 6.80 (d, *J* = 2.3 Hz, 1 H), 4.63 (s, 1 H), 4.16 (q, *J* = 7.1 Hz, 2 H), 3.03 (s, 3 H), 1.19 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 165.05 (C), 159.70 (C), 139.62 (C), 134.43 (C), 130.77 (C), 119.82 (CH), 117.37 (CH), 113.11 (CH), 64.54 (CH), 62.65 (CH₂), 36.48 (CH₃), 13.99 (CH₃).

HRMS (ESI⁺): *m/z* = 270.0517 [M + H]⁺, C₁₂H₁₃ClNO₄ requires 270.0533.

Ethyl 4,7-Dimethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4da)

Yield: 25.6 mg (68%); brown oil.

IR (neat): 1755, 1740, 1506, 1213, 1178, 1043, 699 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 6.91 (ddd, *J* = 8.1, 1.8, 0.8 Hz, 1 H), 6.86 (dd, *J* = 1.9, 0.7 Hz, 1 H), 6.71 (d, *J* = 8.1 Hz, 1 H), 4.61 (s, 1 H), 4.13 (q, *J* = 7.1 Hz, 2 H), 3.02 (s, 3 H), 2.28 (s, 3 H), 1.16 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 165.6 (C), 160.8 (C), 141.0 (C), 131.0 (C), 130.0 (C), 125.9 (CH), 117.0 (CH), 112.7 (CH), 65.0 (CH₃), 62.3 (CH₂), 36.4 (CH), 20.5 (CH₃), 14.0 (CH₃).

HRMS (ESI⁺): *m/z* = 250.1077 [M + H]⁺, C₁₃H₁₆NO₄ requires 250.1079.

Ethyl 7-Chloro-4-methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4ea)

Yield: 20.7 mg (51%); red oil.

IR (neat): 1769, 1734, 1500, 1195, 1013, 956, 850, 802, 690 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 7.08 (dt, *J* = 5.6, 2.2 Hz, 2 H), 6.74 (d, *J* = 8.5 Hz, 1 H), 4.64 (s, 1 H), 4.15 (q, *J* = 7.1 Hz, 2 H), 3.03 (s, 3 H), 1.18 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 165.09 (C), 159.75 (C), 141.22 (C), 132.29 (C), 125.36 (C), 124.84 (CH), 116.83 (CH), 113.62 (CH), 64.66 (CH₂), 62.59 (CH), 36.53 (CH₃), 13.99 (CH₃).

HRMS (ESI⁺): *m/z* = 270.0523 [M + H]⁺, C₁₂H₁₃ClNO₄ requires 270.0533.

Ethyl 4-Ethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4ab)

Yield: 25.8 mg (69%); brown oil.

IR (neat): 1770, 1674, 1499, 1266, 1206, 1017, 747 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 7.10 (ddd, *J* = 8.4, 7.2, 1.5 Hz, 1 H), 7.10–7.00 (m, 1 H), 6.93–6.81 (m, 2 H), 4.77 (s, 1 H), 4.11 (q, *J* = 7.1 Hz, 2 H), 3.60 (dq, *J* = 14.2, 7.1 Hz, 1 H), 3.29 (dq, *J* = 14.4, 7.2 Hz, 1 H), 1.29 (t, *J* = 7.2 Hz, 3 H), 1.14 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 165.99 (C), 161.00 (C), 141.35 (C), 132.83 (C), 125.37 (CH), 120.03 (CH), 116.69 (CH), 113.34 (CH), 62.29 (CH), 62.17 (CH₂), 43.39 (CH₂), 13.91 (CH₃), 12.30 (CH₃).

HRMS (ESI⁺): *m/z* = 250.1067 [M + H]⁺, C₁₃H₁₆NO₄ requires 250.1079.

Ethyl 6-Chloro-4-ethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4cb)

Yield: 24.3 mg (57%); red oil.

IR (neat): 1778, 1734, 1501, 1189, 1131, 1046, 836, 804 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 6.97 (dd, *J* = 8.1, 0.8 Hz, 1 H), 6.89–6.80 (m, 2 H), 4.77 (s, 1 H), 4.15 (q, *J* = 7.1 Hz, 2 H), 3.55 (dq, *J* = 14.2, 7.1 Hz, 1 H), 3.28 (dq, *J* = 13.6, 7.3 Hz, 1 H), 1.29 (t, *J* = 7.2 Hz, 3 H), 1.18 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 165.6 (C), 160.2 (C), 139.8 (C), 133.8 (C), 130.7 (C), 119.7 (CH), 117.6 (CH), 113.4 (CH), 62.6 (CH₂), 61.8 (CH), 43.5 (CH₂), 14.0 (CH₃), 12.2 (CH₃).

HRMS (ESI⁺): *m/z* = 284.0689 [M + H]⁺, C₁₃H₁₅ClNO₄ requires 284.0684.

Ethyl 4-Ethyl-7-methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4db)

Yield: 26.9 mg (68%); red oil.

IR (neat): 1770, 1679, 1512, 1295, 1199, 1150, 1021, 801 cm⁻¹.

¹H NMR (300 MHz, CDCl₃): δ = 6.91 (ddd, *J* = 8.1, 1.8, 0.7 Hz, 1 H), 6.87 (d, *J* = 1.3 Hz, 1 H), 6.77 (d, *J* = 8.1 Hz, 1 H), 4.73 (s, 1 H), 4.11 (q, *J* = 7.1 Hz, 2 H), 3.57 (dq, *J* = 14.2, 7.1 Hz, 1 H), 3.25 (dq, *J* = 14.4, 7.2 Hz, 1 H), 2.28 (s, 3 H), 1.27 (t, *J* = 7.2 Hz, 4 H), 1.15 (t, *J* = 7.1 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 166.14 (C), 161.30 (C), 141.31 (C), 130.39 (C), 130.02 (C), 125.84 (CH), 117.22 (CH), 113.29 (CH), 62.29 (CH₂), 62.23 (CH), 43.45 (CH), 20.47 (CH₃), 13.97 (CH₃), 12.36 (CH₃).

HRMS (ESI⁺): *m/z* = 264.1229 [M + H]⁺, C₁₄H₁₈NO₄ requires 264.1236.

N-Alkylation/Transesterification of Benzoxazinone; Methyl 4-Methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (4aa')

To a solution of benzoxazinone **2a** (0.15 mmol) in CH_2Cl_2 (2 mL) at 0 °C was added dropwise a 2 M Me_2Zn solution in toluene (**3a**; 0.3 mmol) under N_2 atmosphere. After the addition, the reaction mixture was placed at r.t. and stirred for 30 min. Anhyd MeOH (3 mmol) was added to the mixture and the resulting mixture was stirred till completion of the reaction (TLC, $\pm$ 30 min). The reaction was quenched with a solution of Rochelle salt (5 mL), extracted with CH_2Cl_2 (3×15 mL), the combined organic layers were dried (anhyd MgSO_4), and concentrated under reduced pressure. The resulting product was purified by flash column chromatography on silica gel (hexane/EtOAc) to afford the product **4aa'**; yield: 18.6 mg (56%); brown oil.

IR (neat): 1773, 1726, 1504, 1215, 1195, 1023, 740 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.13 (td, J = 7.7, 1.5 Hz, 1 H), 7.05 (dd, J = 8.0, 1.5 Hz, 1 H), 6.97–5.88 (m, 2 H), 4.69 (s, 1 H), 3.69 (s, 3 H), 3.05 (s, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 166.0 (C), 160.3 (C), 141.0 (C), 133.3 (C), 125.6 (CH), 120.2 (CH), 116.5 (CH), 113.0 (CH), 64.8 (CH), 53.0 (CH_3), 36.4 (CH_3).

HRMS (ESI⁺): m/z = 222.0760 $[\text{M} + \text{H}]^+$, $\text{C}_{11}\text{H}_{12}\text{N}_4\text{O}_4$ requires 222.0766.

Nucleophilic N-Alkylation of Quinoxalinones; General Procedure

To a solution of quinoxalinone **6** (0.15 mmol) in CH_2Cl_2 (2 mL) at 0 °C was added dropwise a 1 M Et_2Zn solution in hexanes (**3b**; 0.3 mmol) under N_2 atmosphere. After the addition, the reaction mixture was placed at r.t. and stirred till completion of the reaction (TLC, $\pm$ 30 min). The reaction was quenched with a solution of Rochelle salt (5 mL), extracted with CH_2Cl_2 (3×15 mL), the combined organic layers were dried (anhyd MgSO_4), and concentrated under reduced pressure. The resulting product was purified by flash column chromatography on silica gel (hexane/EtOAc) to afford the desired product **7**.

Ethyl 1-Ethyl-4-methyl-3-oxo-1,2,3,4-tetrahydroquinoxaline-2-carboxylate (7ab)

Yield: 27.9 mg (71%); yellow oil.

IR (neat): 1699, 1390, 1242, 1148, 764, 663 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.12–7.01 (m, 1 H), 6.96 (dd, J = 8.0, 1.5 Hz, 1 H), 6.90–6.79 (m, 2 H), 4.67 (s, 1 H), 4.09 (qd, J = 7.1, 2.8 Hz, 2 H), 3.58 (dd, J = 13.7, 6.9 Hz, 1 H), 3.42 (s, 3 H), 3.26 (dd, J = 13.6, 7.2 Hz, 1 H), 1.27 (t, J = 7.2 Hz, 3 H), 1.15 (t, J = 7.1 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 167.89 (C), 162.17 (C), 135.36 (C), 129.24 (C), 124.06 (CH), 119.24 (CH), 114.90 (CH), 112.54 (CH), 64.40 (CH), 61.65 (CH_2), 43.64 (CH_2), 29.47 (CH_3), 14.06 (CH_3), 12.34 (CH_3).

HRMS (ESI⁺): m/z = 263.1389 $[\text{M} + \text{H}]^+$, $\text{C}_{14}\text{H}_{19}\text{N}_2\text{O}_3$ requires 263.1396.

Ethyl 4-Benzyl-1-ethyl-3-oxo-1,2,3,4-tetrahydroquinoxaline-2-carboxylate (7bb)

Yield: 42.6 mg (84%); clear oil.

IR (neat): 1727, 1671, 1506, 1405, 1200, 1004, 741, 720, 696 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.60–7.14 (m, 5 H), 7.01 (ddd, J = 8.2, 7.3, 1.5 Hz, 1 H), 6.84 (dt, J = 8.0, 1.4 Hz, 2 H), 6.73 (ddd, J = 8.2, 7.3, 1.3 Hz, 1 H), 5.75–5.50 (m, 1 H), 4.77 (s, 1 H), 4.73 (d, J = 16.2 Hz, 1 H), 4.14 (qd, J = 7.1, 0.8 Hz, 2 H), 3.63 (dq, J = 14.1, 7.1 Hz, 1 H), 3.35 (dq, J = 13.5, 7.2 Hz, 1 H), 1.31 (t, J = 7.2 Hz, 3 H), 1.18 (t, J = 7.1 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 167.90 (C), 162.30 (C), 136.58 (C), 135.38 (C), 128.81 (C), 128.69 (CH), 127.14 (CH), 126.25 (CH), 124.14 (CH), 119.39 (CH), 115.74 (CH), 112.79 (CH), 64.33 (CH), 61.69 (CH_2), 46.51 (CH_2), 43.82 (CH_2), 14.07 (CH_3), 12.46 (CH_3).

HRMS (ESI⁺): m/z = 339.1704 $[\text{M} + \text{H}]^+$, $\text{C}_{20}\text{H}_{23}\text{N}_2\text{O}_3$ requires 339.1709.

Ethyl 1-Ethyl-4-(2-methoxy-2-oxoethyl)-3-oxo-1,2,3,4-tetrahydroquinoxaline-2-carboxylate (7cb)

Yield: 36.1 mg (75%); yellow oil.

IR (neat): 1733, 1674, 1396, 1289, 1216, 1194, 1012, 744 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.15–6.98 (m, 1 H), 6.91–6.77 (m, 2 H), 6.72 (dd, J = 8.0, 1.3 Hz, 1 H), 4.96 (d, J = 17.5 Hz, 1 H), 4.71 (s, 1 H), 4.45 (d, J = 17.5 Hz, 1 H), 4.11 (qd, J = 7.1, 1.1 Hz, 2 H), 3.75 (s, 3 H), 3.59 (dt, J = 14.1, 7.0 Hz, 1 H), 3.30 (dq, J = 14.4, 7.2 Hz, 1 H), 1.27 (t, J = 7.2 Hz, 3 H), 1.15 (t, J = 7.1 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 168.35 (C), 167.36 (C), 162.27 (C), 135.42 (C), 128.16 (C), 124.40 (CH), 119.41 (CH), 114.39 (CH), 113.09 (CH), 64.12 (CH), 61.72 (CH_2), 52.44 (CH_3), 44.01 (CH_2), 43.82 (CH_2), 13.97 (CH_3), 12.35 (CH_3).

HRMS (ESI⁺): m/z = 321.1450 $[\text{M} + \text{H}]^+$, $\text{C}_{16}\text{H}_{21}\text{N}_2\text{O}_5$ requires 321.1451.

N-Alkylation/C-Alkylation of Benzoxazinones; General Procedure

To a solution of benzoxazinone **2** (0.15 mmol) in CH_2Cl_2 (2 mL) at 0 °C was added dropwise a 2 M Me_2Zn solution in toluene/1 M Et_2Zn solution in hexanes **3** (0.3 mmol) under N_2 atmosphere. After the addition, the reaction mixture was placed at r.t. and stirred for 30 min. After 30 min, the appropriate electrophile (0.75 mmol) was introduced in the reaction, which was stirred at r.t. till completion of the reaction (TLC). The reaction was quenched with a solution of Rochelle salt (5 mL), extracted with CH_2Cl_2 (3×15 mL), the combined organic layers were dried (anhyd MgSO_4), and concentrated under reduced pressure. The resulting product was purified by flash column chromatography on silica gel (hexane/EtOAc) to afford the product **8a–h**.

Ethyl 3-Benzyl-4-methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8a)

Yield: 18.1 mg (37%); yellow oil.

IR (neat): 1757, 1738, 1499, 1272, 1218, 1181, 1018, 742, 698 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.21–7.09 (m, 5 H), 7.00 (ddd, J = 8.0, 7.5, 1.5 Hz, 1 H), 6.81 (dd, J = 7.9, 1.6 Hz, 1 H), 6.74–6.59 (m, 2 H), 4.32–4.13 (m, 2 H), 3.72 (d, J = 15.0 Hz, 1 H), 3.50 (d, J = 15.0 Hz, 1 H), 2.89 (s, 3 H), 1.17 (t, J = 7.1 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 167.43 (C), 163.76 (C), 139.57 (C), 134.09 (C), 133.01 (C), 130.00 (CH), 128.27 (CH), 127.14 (CH), 125.36 (CH), 119.03 (CH), 116.03 (CH), 111.94 (CH), 71.79 (C), 62.58 (CH_2), 37.73 (CH_2), 32.48 (CH_3), 13.96 (CH_3).

HRMS (ESI⁺): m/z = 326.1389 $[\text{M} + \text{H}]^+$, $\text{C}_{19}\text{H}_{20}\text{N}_2\text{O}_4$ requires 326.1392.

Ethyl 3-Benzyl-4-ethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8b)

Yield: 26.0 mg (51%); yellow oil.

IR (neat): 1753, 1736, 1499, 1235, 1203, 1045, 735, 697 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.12–7.08 (m, 5 H), 6.98 (ddd, J = 8.1, 7.4, 1.6 Hz, 1 H), 6.79 (dd, J = 8.0, 1.6 Hz, 1 H), 6.72–6.58 (m, 2 H), 4.27 (qd, J = 7.1, 1.7 Hz, 2 H), 3.62 (d, J = 14.8 Hz, 1 H), 3.51–3.25 (m, 2 H), 3.19–3.02 (m, 1 H), 1.24 (t, J = 7.1 Hz, 3 H), 1.07 (t, J = 7.0 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 168.2 (C), 164.1 (C), 139.6 (C), 133.9 (C), 131.4 (CH), 130.3 (C), 128.2 (CH), 127.1 (CH), 125.3 (CH), 118.6 (CH), 116.3 (CH), 112.1 (CH), 72.3 (C), 62.6 (CH_2), 40.8 (CH_2), 38.6 (CH_2), 13.9 (CH_3), 12.3 (CH_3).

HRMS (ESI⁺): m/z = 340.1548 $[\text{M} + \text{H}]^+$, $\text{C}_{20}\text{H}_{22}\text{NO}_4$ requires 340.1549.

Ethyl 3-Benzyl-4,6-dimethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8c)

Yield: 35.5 mg (67%); clear oil.

IR (neat): 1755, 1740, 1506, 1234, 1213, 1078, 1043, 799, 699 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.41–7.09 (m, 5 H), 6.69 (d, J = 8.0 Hz, 1 H), 6.52–6.40 (m, 2 H), 4.25 (qd, J = 7.2, 1.9 Hz, 2 H), 3.63 (d, J = 14.8 Hz, 1 H), 3.43 (d, J = 14.7 Hz, 1 H), 3.47–3.27 (m, 1 H), 3.09 (dq, J = 14.2, 7.1 Hz, 1 H), 1.23 (t, J = 7.1 Hz, 3 H), 1.06 (t, J = 7.0 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 168.3 (C), 164.2 (C), 137.7 (C), 134.9 (C), 134.1 (C), 131.1 (C), 130.3 (CH), 128.1 (CH), 127.1 (CH), 119.2 (CH), 116.0 (CH), 112.8 (CH), 72.3 (C), 62.6 (CH_2), 40.8 (CH_2), 38.5 (CH_2), 21.4 (CH_3), 13.9 (CH_3), 12.42 (CH_3).

HRMS (ESI⁺): m/z = 354.1709 $[\text{M} + \text{H}]^+$, $\text{C}_{21}\text{H}_{24}\text{NO}_4$ requires 354.1705.

Ethyl 3-Benzyl-7-chloro-4-methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8d)

Yield: 40.4 mg (72%); yellow oil.

IR (neat): 1763, 1744, 1497, 1264, 1240, 1042, 1021, 795, 746, 697 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.16–7.01 (m, 5 H), 6.94 (dd, J = 8.7, 2.4 Hz, 1 H), 6.77 (d, J = 2.4 Hz, 1 H), 6.52 (d, J = 8.8 Hz, 1 H), 4.28 (qd, J = 7.1, 3.1 Hz, 2 H), 3.61 (d, J = 14.7 Hz, 1 H), 3.46 (s, 1 H), 3.47–3.28 (m, 1 H), 3.10 (dq, J = 14.3, 7.1 Hz, 1 H), 1.25 (t, J = 7.1 Hz, 3 H), 1.06 (t, J = 7.0 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 167.8 (C), 163.6 (C), 139.6 (C), 133.6 (C), 130.3 (C), 130.2 (CH), 128.3 (CH), 127.3 (CH), 125.0 (CH), 123.0 (C), 116.6 (CH), 112.7 (CH), 72.2 (C), 62.8 (CH_2), 40.9 (CH_2), 38.8 (CH_2), 13.9 (CH_3), 12.1 (CH_3).

HRMS (ESI⁺): m/z = 374.1165 $[\text{M} + \text{H}]^+$, $\text{C}_{20}\text{H}_{21}\text{ClNO}_4$ requires 374.1159.

Ethyl 3-Allyl-4-ethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8e)

Yield: 27.3 mg (63%); brown oil.

IR (neat): 1762, 1739, 1498, 1271, 1215, 1130, 1020, 744, 660 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.12–7.07 (m, 1 H), 7.06–6.99 (m, 1 H), 6.85 (dd, J = 7.7, 1.6 Hz, 1 H), 6.85–6.75 (m, 1 H), 4.12 (qd, J = 7.1, 1.4 Hz, 1 H), 3.45 (dq, J = 14.4, 7.2 Hz, 2 H), 3.28 (dd, J = 14.9, 7.0 Hz, 1 H), 3.26–3.11 (m, 1 H), 3.00–2.85 (m, 1 H), 1.20 (t, J = 7.0 Hz, 3 H), 1.10 (t, J = 7.1 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 168.2 (C), 163.6 (C), 140.6 (C), 132.5 (C), 131.1 (CH), 125.3 (CH), 120.2 (CH_2), 119.6 (CH), 116.6 (CH), 113.4 (CH), 70.8 (C), 62.3 (CH_2), 40.1 (CH_2), 36.9 (CH_2), 13.9 (CH_3), 13.0 (CH_3).

HRMS (ESI⁺): m/z = 290.1385 $[\text{M} + \text{H}]^+$, $\text{C}_{16}\text{H}_{20}\text{NO}_4$ requires 290.1392.

Ethyl 3-Allyl-4-ethyl-6-methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8f)

Yield: 30.5 mg (67%); red oil.

IR (neat): 1761, 1739, 1506, 1277, 1215, 1180, 1077, 925, 802 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 6.89 (d, J = 8.0 Hz, 1 H), 6.63 (ddd, J = 8.0, 1.8, 0.7 Hz, 1 H), 6.63–6.54 (m, 1 H), 5.87–5.69 (m, 1 H), 5.22 (dq, J = 17.1, 1.6 Hz, 1 H), 5.15 (dq, J = 10.2, 1.4 Hz, 1 H), 4.13 (q, J = 7.1 Hz, 2 H), 3.43 (dq, J = 14.4, 7.1 Hz, 1 H), 3.34–3.22 (m, 1 H), 3.17 (ddt, J = 15.1, 6.4, 1.6 Hz, 1 H), 2.90 (ddt, J = 15.1, 7.5, 1.2 Hz, 1 H), 2.32 (s, 3 H), 1.19 (t, J = 7.0 Hz, 3 H), 1.12 (t, J = 7.1 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 168.3 (C), 163.6 (C), 138.5 (C), 135.0 (C), 132.1 (C), 131.2 (CH), 120.1 (CH), 120.1 (CH_2), 116.2 (CH), 114.0 (CH), 70.8 (C), 62.3 (CH_2), 40.1 (CH_2), 36.9 (CH_2), 21.4 (CH_3), 13.9 (CH_3), 13.0 (CH_3).

HRMS (ESI⁺): m/z = 304.1550 $[\text{M} + \text{H}]^+$, $\text{C}_{17}\text{H}_{22}\text{NO}_4$ requires 304.1549.

Ethyl 3-Allyl-4-ethyl-7-methyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8g)

Yield: 30.9 mg (68%); orange oil.

IR (neat): 1762, 1738, 1511, 1273, 1229, 1144, 1020, 802 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 6.88 (ddd, J = 8.2, 1.9, 0.8 Hz, 1 H), 6.84 (dd, J = 1.8, 0.8 Hz, 1 H), 6.70 (d, J = 8.3 Hz, 1 H), 5.90–5.70 (m, 1 H), 5.22 (dq, J = 17.1, 1.6 Hz, 1 H), 5.16 (dq, J = 10.2, 1.4 Hz, 1 H), 4.11 (qd, J = 7.2, 0.9 Hz, 2 H), 3.42 (dq, J = 14.4, 7.2 Hz, 1 H), 3.32–3.21 (m, 1 H), 3.22–3.12 (m, 1 H), 2.90 (ddt, J = 15.1, 7.4, 1.2 Hz, 1 H), 2.27 (s, 3 H), 1.17 (t, J = 7.0 Hz, 3 H), 1.10 (t, J = 7.1 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 168.3 (C), 163.8 (C), 140.6 (C), 131.3 (CH), 130.1 (C), 129.6 (C), 125.7 (CH), 120.0 (CH_2), 117.1 (CH), 113.5 (CH), 70.8 (C), 62.2 (CH_2), 40.1 (CH_2), 36.7 (CH_2), 20.4 (CH_3), 13.9 (CH_3), 13.1 (CH_3).

HRMS (ESI⁺): m/z = 304.1550 $[\text{M} + \text{H}]^+$, $\text{C}_{17}\text{H}_{22}\text{NO}_4$ requires 304.1549.

Ethyl 3-Allyl-7-chloro-4-ethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8h)

Yield: 24.8 mg (51%); orange oil.

IR (neat): 1770, 1740, 1494, 1273, 1218, 1147, 1077, 858, 801 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.08–7.01 (m, 2 H), 6.71 (d, J = 8.5 Hz, 1 H), 5.76 (dddd, J = 16.7, 10.2, 7.5, 6.4 Hz, 1 H), 5.29–5.13 (m, 2 H), 4.13 (q, J = 7.1 Hz, 2 H), 3.43 (dq, J = 14.4, 7.2 Hz, 1 H), 3.33–3.23 (m, 1 H), 3.18 (ddt, J = 15.2, 6.4, 1.4 Hz, 1 H), 2.92 (ddt, J = 15.1, 7.4, 1.1 Hz, 1 H), 1.18 (t, J = 6.9 Hz, 3 H), 1.13 (t, J = 7.2 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 167.8 (C), 162.9 (C), 140.7 (C), 131.4 (C), 130.8 (CH), 125.1 (CH), 124.2 (C), 120.5 (CH_2), 116.9 (CH), 114.1 (CH), 70.7 (C), 62.5 (CH_2), 40.2 (CH_2), 36.9 (CH_2), 13.9 (CH_3), 12.78 (CH_3).

HRMS (ESI⁺): m/z = 324.1003 $[\text{M} + \text{H}]^+$, $\text{C}_{16}\text{H}_{19}\text{ClNO}_4$ requires 324.1003.

Ethyl (E)-3-(4-Ethoxy-4-oxobut-2-en-1-yl)-4-ethyl-2-oxo-3,4-dihydro-2H-benzo[b][1,4]oxazine-3-carboxylate (8i)

To a solution of **4ab** (0.1 mmol) in THF (1 mL) at -78°C was added a 2 M solution of LDA in THF (1.2 mmol) and the reaction mixture was stirred for 0.5 h at -78°C . After 0.5 h, ethyl (E)-4-iodobut-2-enoate was added at -78°C and the mixture was allowed to reach r.t. The mixture was stirred till completion of the reaction (TLC). The reaction was quenched with sat. aq. NH_4Cl (5 mL), extracted with CH_2Cl_2 (3 $\times$ 15 mL), the combined organic layers were dried (anhyd. MgSO_4), and concentrated under reduced pressure. The resulting product was purified by flash column chromatography on silica gel (hexane/EtOAc) to afford the product **8i**; yield: 20.1 mg (37%); orange oil.

IR (neat): 1763, 1730, 1716, 1497, 1263, 1215, 1187, 1130, 1092, 1037, 746 cm^{-1} .

^1H NMR (300 MHz, CDCl_3): δ = 7.10 (ddd, J = 8.1, 7.4, 1.6 Hz, 1 H), 7.06–7.02 (m, 1 H), 6.94–6.71 (m, 3 H), 5.98 (dt, J = 15.7, 1.5 Hz, 1 H), 4.24–4.05 (m, 4 H), 3.54–3.18 (m, 3 H), 3.06 (ddd, J = 15.6, 7.8, 1.4 Hz, 1 H), 1.30–1.16 (m, 6 H), 1.09 (t, J = 7.1 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 167.7 (C), 165.6 (C), 163.2 (C), 140.7 (CH), 140.7 (C), 132.2 (C), 126.0 (CH), 125.5 (CH), 120.1 (CH), 116.7 (CH), 113.7 (CH), 70.3 (C), 62.6 (CH_2), 60.5 (CH_2), 40.3 (CH_2), 35.0 (CH_2), 14.2 (CH_3), 13.8 (CH_3), 12.9 (CH_3).

HRMS (ESI^+): m/z = 362.1587 [$\text{M} + \text{H}$] $^+$, $\text{C}_{19}\text{H}_{24}\text{NO}_6$ requires 362.1598.

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Supporting Information

Supporting information for this article is available online at <http://dx.doi.org/10.1055/s-0036-1588742>.

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